

Institute of Actuaries of India

Subject CA1-II – Actuarial Risk Management

October 2014 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Soln 1:

Data checks that can be included are:

- Reconciliation of
 - total number of members in each category (pensioners, deferred and active members) using the previous period data and movement data
 - total benefit at valuation date against the total benefit at the last valuation
 - total contributions paid in this year against that made in the previous year
 - salary levels between two valuation dates
- Independent check with the trust accounting statements
 - Contribution and benefits paid out from the Valuation data with the corresponding figures in the accounts
 - Salaries given in the valuation date against the salary as per the accounts
 - Pension in payment over the period with the accounts figures
 - Investment income as per the assets data and that shown in the accounts
- Consistency between
 - Salary related contributions and in-payment benefits (say annuity or pension benefits) as indicated by membership data and corresponding figures from the accounts
- Check for unusual values like impossible date of births or start dates, negative salaries or contributions, etc
- Check with previous investigation data for information that should not change at member level like date of birth, etc
- If the assets are held by third party, reconciliation between the records of both the two parties
- Random spot checks on individual records, especially the ones with very large benefit (materiality)

[5 Marks]

Soln. 2:

(i) An Actuary should

- ensure he is familiar with the context in which he is going to operate and implications of the results
- consider the resources and timescales
- define the task with the client and consider conflict of interest
- establish what are the questions that require answering
- gather and assess the available information
- set assumptions
- decide on a method
- arrive at “the solution”
- check the solution
- communicate the answer in a way that is understood by the client
- consider the professional implications of the work being done

[4]

(ii) Principles to be followed while managing conflict of interest are:

- Avoid the conflict, by declining the assignment
- Chinese wall, ring fencing people and data
- Disclosure of the conflict to both parties
- Keeping detailed records of work assignments

[2]

[6 Marks]

Solution 3:**(i) Investigations required for setting reinsurance retention limit**

- Retention limit should be such that it balances the advantage of reducing the risk volatility and the disadvantage of passing on the profits to the reinsurer
- Determining the appropriate retention requires modelling, one should start with:
 - Estimated claims cost that the reinsurer will pay over the term of the reinsurance for a given retention limit
 - This value will need scenario testing and it would be sensitive to the claims distribution assumed
- In the long run, the cost of reinsurance claims would be lower than the reinsurance outgo (reinsurance premium), as the reinsurance premium should have loading for profit and expenses of the reinsurer
- However, as explained before; reinsurance brings stability to the profits

Actual modelling requires:

- A stochastic model needs to be adopted with mortality being modeled stochastically
- Since mortality will be the main risk transferred to the reinsurer, it makes sense to model that as a stochastic value
- Need to set criteria for testing the probability of ruin or any other profit criteria
- Project the balance sheet and P&L account for a certain period (5 years or 10 years)
- Run at least 10,000 simulations for a given retention
- Note down the output of solvency position of the company
- Run the model for different retention limits
- Analyze the output parameters like mean variance, etc
- Sensitivity analysis may be required for varying the other parameters in the model (not stochastically modeled)

Do the exercise till you don't get a retention limit that best meets all the criteria.

[6]

(ii) Other factors to be taken into account

- The available free assets will give the margin for withstanding adverse claims experience
- Management's attitude towards risk; would be driven by the free assets available also
- What probability of insolvency should be taken for testing
- How much credit does the regulations provide in solvency due to reinsurance; if there is a cap in the credit that can be taken, say maximum reduction in reserve allowed is 50%
- How important is it for the management to have stable profits; less retention more stable the profits
- What is the expected volume of business; and what kind of policies; very high ticket size; what mix of business are we expecting
- What are the competitors doing; what is their retention limits and amount of business reinsured
- Reinsurer's strength
 - What is the credit rating of the reinsurer
 - What kind of technical assistance can the reinsurer provide; is there a minimum amount of business to be reinsured to get this technical support
 - How will the market react to the change in reinsurance program
- The arrangement should be simple and practical to implement

[4]

[10 Marks]

Soln. 4:

i)

- “Price and yield are inversely related”, this statement is true.
- From the formula for price of a bond you can justify the same.
- So when the price of a bond increases its gross redemption yield will fall.
- However when you buy a bond for a certain price you are locking into the GRY implied by that price.
- So as long as you hold the bond to maturity you will get that yield, irrespective of the market movements in the price.
- Yield to maturity will be the same as your calculation while buying the bond, except for some second order differences like:
 - Reinvestment of coupons at a different rate
 - Tax risk, default risk
 - Currency risk in case of overseas investment
- If you sell the bond prior to redemption, then you could either make a gain or losses depending upon if you are selling the bond at a lower yield or higher yield.

[4]

ii)

- The primeness of the property will decide the risk involved and hence the discount rate for valuation.
- More risky the asset, higher the discount rate
- The level of risk of a property can be broadly understood in:
 - Default risk
 - Void risk
 - Volatility of returns risk
 - Marketability risk
- All these risks can be assessed based on certain proxies like
 - Age of the building
 - Location – good locality will mean less risk, lower discount rate
 - Usage - shops, residential, office, factory, etc
 - Can it be used for alternate purposes; the more flexible the use of the building; the better or lower risk
 - Quality of the building
 - Is it in good condition
 - Design and access to the building
 - Is there any developmental potential
 - Terms of the lease – duration, rent reviews, whether full insurance
 - Will they do the maintenance and repairing
 - Single occupation or multiple occupation
 - Financial strength of the occupant/s
 - Prospect of rental growth
 - Also over supply in the market for the same kind of property
 - Condition of the local economy

[9]**[13 Marks]**

Soln. 5:**(i)** Marketing promotion can help the company:

- To increase company/product visibility in the market
- To promote its brand name
- To be a differentiator in the market
- To create a strategy to attract customer's interest being the offer is of free in nature
- To make an alternative to direct advertisement
- To increase the awareness and to have several personal details such as contact number and addresses which may be of used for further new business sale.

[3]**(ii)** Will need to estimate the number of policies being taken on by:

- Estimate the expected number of new cars that could be bought by your customers- will need looking socio-economic profile
- Estimate the number of accidental deaths/ disability within one year of buying
- Consider the length of the promotional period.
- Estimate the type/make/model of cars that will be bought by your customers- as it could impact frequency of accidents.

[3]**(iii)** The main issue is that the company has taken on the risk but with no premium in which to offset it.

- It will need to actively manage risks by monitoring the number of contracts issued and the numbers of death/ disability.
- Consider diversifying the risk away as part of a broader portfolio.
- Could use reinsurance to transfer some of the risk.
- This offer should available on only motor vehicles of value well higher than a certain amount.
- Consider restrictions on the offer, for example certain models of vehicles.
- Ensure that the claims control procedures work, i.e. underwriting at point of claim.
- Implement control systems to reduce operational risks such as financial fraud.
- Must also monitor the expenses associated with these policies.
- Need to determine an appropriate amount of capital to hold against the risks accepted.
- In determining the appropriate amount of capital will need to determine the expected cost. This will be based on cost estimates and use sensitivities to consider what the worst case scenarios could be.
- This could be expressed as a ruin probability over the year.
- Sensitivities will be in relation to key parameters, for example number of car sales or accident rate or mortality.
- Need to consider length of the promotional period or how it will be financed in the long term.
- For example if it will be financed by the profits the general insurer might generate from sales of other products.
- If product is being used to generate sales elsewhere will need to monitor the level of sales generated. Will also need to ensure risks from these sales are actively managed, to ensure these lead to profits.
- Care as to terms and conditions offered in policy contract and designing appropriate exclusions and lowering moral hazard.

[7]**[13 Marks]**

Soln. 6:**(i)**

The Sales Head's suggestion is valid and the below could be some of the supporting arguments for there to be no need, up to a certain level of cover, for individual evidence of insurability:

- First, employees are generally healthier than the general population since they must be fit enough to work full-time.
- Also, some employers require health screenings before offering permanent employment and would less likely employ ill or disabled lives.
- Furthermore, there is less anti-selection since employees normally cannot choose the amount of basic insurance (the basic coverage amount is pre-determined per employee) and also because individuals would not normally apply for employment at a specific company for the express purpose of obtaining insurance, especially as there are barriers to becoming employed.
- If all employees (in a compulsory plan) or a certain minimum percentage of employees (in a voluntary plan) are covered by insurance means that there is a good spread of the mortality risk.
- It is well known that roughly 95% of individual insurance applicants are accepted without substandard ratings, and that in effect the entire cost of individual underwriting is to catch that 5% of applicants who are then declined or rated.
- In the context of group insurance, that 5% of individual applicants would either be unfit to work (and therefore not eligible to be covered) or, for those who are working, the extra mortality risk they represent can then be spread over the remaining lives of the group.

[6]

(ii) The factors to consider while designing the product to reduce the risk of anti-selection are:

- Insurance must be incidental to the existence of the group. This is clearly the case for employer-employee relationships, creditor/borrower groups
- The determination of benefits per member should not be at the discretion of the employer or employee. That is, there should be an automatic basis to determine the level of coverage per employee.
- The objective is to avoid anti-selection by less healthy employees who are expected to choose higher levels of coverage.
- The level of cover may be the same for all employees or may be a function of employee rank (e.g. worker, manager, executive), salary, years of employment, or a combination of these.
- Various scheme designs can be discussed with employers to determine the more suitable option. If benefits were linked to a variable, for example salary or years of employment, an annual re-determination of the insurance cover would have to be carried out.
- Only permanent full time employees should be eligible for insurance cover.
- The employee should be actively at work on the date he or she becomes eligible for insurance.
- Furthermore, extra eligibility conditions may apply such as not having been absent from work due to sickness for more than 3 weeks per year during the previous 2 years (that is, the employee should have been working the normal hours required by the employer).
- Certain age restrictions may also apply, requiring that the employee be younger than age 60 and have joined the employer before attaining age 55.
- New employees must normally satisfy a waiting period, usually a month, before becoming eligible for insurance.
- In a voluntary plan, once they have satisfied this probationary period, they must join the scheme within 30 days or otherwise have to provide some evidence of insurability.
- There may be instances when the employer continues to pay the employee even if he or she is temporarily off the job. It is important for the insurer to clarify instances where this is allowed

and for what duration.

- Also, the employer should not employ its own discretion as to whether to continue paying an employee who is not actively at work, but should follow an established policy that is applied to all employees or certain classes of employees.
- In the end, there should be no ambiguity about whether the insurance coverage is in force when the employee is not actively at work, for example as a result of illness or maternal leave.

[7]

[13 Marks]

Soln. 7:

(i) The six methods of financing defined benefit pension schemes are:

1. Pay as you go:

- It is an unfunded method of financing
- Benefits are provided for as they become due to be paid, without setting them aside before
- Many government schemes may be financed by this method.

2. Lumpsum in advance:

- Funds are expected to be sufficient to meet the cost of the benefit as soon as the benefit promise is made
- It is like paying a single premium at entry or the moment the member is eligible to join the scheme

3. Terminal funding:

- Funds are expected to be sufficient to meet the cost of a series of benefit payments as soon as the first tranche becomes payable

4. Regular contributions:

- Funds are gradually built up to a level expected to be sufficient to meet the cost of the benefit over the period between the promise being made and the benefit first becoming payable.

5. Just in time funding:

- Funds that are expected to be sufficient to meet the cost of the benefit can be set up as soon as a risk arises in relation to the future financing of the benefits (eg bankruptcy or change in control)

6. Smoothed Pay as you go:

- Funds are set up to smooth the costs under a pay as you go approach to allow for the effects of timing differences between contributions and benefits, short term business cycles and long term population change.

[6]

(ii)

(a)

- **Security of the benefit promises**

1. Pay as you go:

- As there is no prior funding, the security of the benefit promise is as good as the financial well-being of the benefit provider.
- From a funding perspective, it has the least security of the benefits promised.

2. Lumpsum in advance:

- As the full value of the benefit is funded when the promise was made, this is the more secure method of funding
- We need to bear in mind that so much advance funding of all future benefit accrual should not cause any financial crisis in the company.

3. Terminal funding:

- As the benefit promised is a single payment at retirement, terminal funding and pay as you go are the same.
- All the comments given for pay as you go applies; least security of benefits promised
- 4. Regular contributions:
 - The security is better as regularly the funding is done.
 - The relative security depends on pace of the regular contributions
- 5. Just in time funding:
 - This method is better than pay as you go for security as it is funded as soon as there is a threat to the security of the employer
 - Under normal circumstances (when employer in sound financial health); the method is similar to pay as you go.
- 6. Smoothed Pay as you go:
 - The security is better than pay as you go, as there is some advance funding
 - The security levels depends on the assumptions used for smoothing

(b) Liquidity of funds

1. Pay as you go:
 - The benefit promised is a lumpsum at retirement, full benefit has to be provided right at the point of retirement.
 - This can be a large amount of money; especially if many people retiring / resigning in the same month
 - Can cause liquidity issue to the employer if many retiring in the same month.
2. Lumpsum in advance:
 - All the future accrual of benefits to be provided at day 1. Based on the number of employees recruited, the liquidity of such a funding can cause strain to the employer
 - And not all may stay till retirement to take the full benefits
3. Terminal funding:
 - The comments are the same as for pay as you go, because the benefit promised is a lumpsum
4. Regular contributions:
 - This is the best funding method from a liquidity perspective, as regular small contributions will not cause much liquidity issue
 - And funding will not be overly sensitive to the number of new recruits or retirees
5. Just in time funding:
 - This can be a problem, as this funding will trigger when there is a suspicion on the financial security of the employer. This might also be a point when the employer will not have liquid assets to fund the benefits, all in one go.
6. Smoothed Pay as you go:
 - Liquidity better than for pay as you go, as there is some smoothing of funding. The sensitivity of number of retirees can be reduced.
 - The liquidity will depend on the assumptions used for the smoothing.

[12]

[18 Marks]

Soln. 8:

(i) Benefits that are normally provided under income protection products are:

- To provide an income to the insured or their dependents on the occurring of the insured event
- Insured event could be long term sickness or incapacity due to accident or illness.
- Insured risk could also be unemployment or
- birth of a handicapped child needing high levels of care
- Income stream would continue till the insured retirement age or a specified shorter period

[2]

(ii) Reasons for excluding explicitly certain causes of disability from the cover of healthcare insurance contracts include:

- To avoid anti-selection – e.g. exclude existing conditions. Exclusion effectively replaces questions on the underwriting form.
- Moral hazard – because of insurance people may be reckless in their behaviour so, for example, disability that results from self-inflicted injury, failure to follow medical advice might be excluded,
- Special risk that is difficult to price – e.g. may exclude disability as a result of participation in a professional sporting event.
- Perils, cost for which is likely to be difficult to price because of lack of data –
- Cost is uncertain in the future – uncertain risk like AIDS, war, terrorism.
- It may be unlawful to provide a benefit, for example, if the policyholder is disabled as a result of committing an illegal act (e.g. committing a terrorist act).
- Payment received from elsewhere – may exclude claims for which the policyholder is already receiving compensation (say from an employer or through a structured settlement).

[6]

(iii) Reporting delays: it is the time between the event happening and the insurance company being informed of it. [

There may be delays due to:

- There would be a natural time-lag between an event and the insurer informed (as it might not be the priority or lack of awareness)
- Delay in compiling required documentation for reporting
- In some cases the conditions emerges over a period; like industrial disease such as asbestosis
- Sometimes it takes time to realize that there is a cause for claiming
- If the amount involved is very small, there is a usual carelessness to claim quickly

Settlement delay: is the time between the claim reported to the insurer and payment of the claim. These delays are due to:

- There will be some time taken to process a claim
- To establish that it is a valid claim and the insurer is liable- as physical condition of claimant may need some time to stabilize and liability of insurer established
- For any investigations to be undertaken; to establish the validity of claim, prevent frauds
- For the condition to stabilize (to know the amount of the loss, for the treatment to be over)
- As part of the policy condition (to wait for disability to be proved as permanent)
- Possible disputes and court settlements

[6]

(iv) Potential reasons for deteriorating experience could be:

- Premium charged is incorrect and not reflective of underlying risks
- If underlying risk profile has changed due to change in target market
- Poor risk selection at the time of initial underwriting
- Poor underwriting /lenient underwriting which caused policies to be issued at standard rates when they would otherwise have been charged sub-standard rates or declined cover
- Changes in target market or sales practices ,i.e, from white collared to blue collared
- Changes of work practices of employers – lower safety standards leading to more accidents at work
- Changes in cover required – the insured event under the income protection policies sold might have changed over time but with limited change in premium income. For eg., Earlier only accidental death/ disability was covered but now even loss of income due to unemployment is also covered; this can change the claims profile over time.
- Anti-selection as rates are more attractive than competition for certain classes of life
- Significant changes in claims personnel, if claim settlement process is more judgemental this could be a problem
- There is a change in the claims processing method or criteria not properly thought through
- over a period, the claims control has got diluted due to market pressures
- Economic changes – if insured event includes unemployment as well; with changing economic environment, the customer behavior to claim would also change
- Benefit inflation – if the income stream is also adjusted for inflation; the inflation over these years were higher than originally estimated
- Claim volatility due to small size of portfolio, law of large nos not working
- Unusual large claims from large policies
- Catastrophic events leading to large claims

Ways to mitigate them:

- Standardization/Improving of the claims processing
- Better initial underwriting and claims control
- Make sure the target market is not changing too much
- The insured events are kept tighter; frequent changes to the wordings can lead to changed customer claiming behavior and claims team getting confused
- It would also lead to too often system change
- Restrict the business to only certain geographic locations; in some locations the propensity to claim could be very different from other places
- Maintain good quality data; it will not reduce the claims experience but will definitely help in monitoring the experience better
- Beyond a point you can't control all the factors; reinsure, as the portfolio is a significant part of the business.

[8]

[22 Marks]
